

SQL Fundamentals: SELECT and Filtering

Exercise 1

1) `SELECT id, first_name, last_name, department,
salary, hire_date, city
FROM employees;`

Result Table:

id	first_name	last_name	department	Salary	hire_date	City
1	Alice	Green	IT	70000	2020-01-10	Johannesburg
2	Brian	Lee	HR	45000	2021-03-20	Cape Town
3	Cathy	Zulu	Finance	65000	2018-07-12	Durban
4	David	Mokoena	Marketing	50000	2021-11-05	Pretoria
5	Eva	Naidoo	IT	72000	2017-07-30	Johannesburg

2) `SELECT DISTINCT department
FROM employees;`

Result Table:

department
IT
HR
Finance
Marketing

3) `SELECT first_name, last_name
FROM employees
ORDER BY salary DESC;`

Result Table:

first_name	last_name
Eva	Naidoo
Alice	Green
Cathy	Zulu
David	Mokoena
Brian	Lee

4) `SELECT id, first-name, last-name, salary
FROM employees
ORDER BY salary DESC LIMIT 3;`
Result Table:

id	first-name	last-name	salary
5	Eva	Naidoo	72000
1	Alice	Green	70000
3	Cathy	Zulu	65000

5) `SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT';`

Result Table:

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT

6) `SELECT id, first-name, last-name, department,
salary
FROM employees
WHERE department = 'Finance' AND salary > 60000;`

Result Table:

id	first-name	last-name	department	salary
3	Cathy	Zulu	Finance	65000

7) `SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'HR' OR department = 'Marketing';`

Result Table:

id	first-name	last-name	department
2	Brian	Lee	HR
4	David	Mokone	Marketing

8) SELECT id, first-name, last-name, department
FROM employees
WHERE department != 'IT';

Result Table

id	first-name	last-name	department
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
4	David	Mokone	Marketing

9) SELECT id, first-name, last-name, department
FROM employees
WHERE department IN ('IT', 'HR', 'Finance');

Result Table:

id	first-name	last-name	department
1	Alice	Green	IT
2	Brian	Lee	HR
3	Cathy	Zulu	Finance
5	Eva	Naidoo	IT

10) SELECT id, first-name, last-name, department
FROM employees
WHERE department = 'IT' AND salary > 6000 AND
city = 'Johannesburg';

Result Table

id	first-name	last-name	department
1	Alice	Green	IT
5	Eva	Naidoo	IT